

Hamza Mounir

Senior Software Engineer • Backend & AI Systems • Cloud Architecture • React

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years designing scalable, cloud-native platforms, microservice architectures, and AI-powered systems. Specialized in Python / FastAPI backends, React & Next.js frontends, and production LLM systems — RAG, vector search, AI agents, and MCP-based tooling. Proven track record taking products from startup launch to enterprise scale in regulated, financial-grade environments, with strong DevOps automation and a focus on reliability, security, and high-quality code delivery.

TECHNICAL SKILLS

Backend & APIs	Python, FastAPI, Node.js, NestJS, REST, gRPC, GraphQL, SQLAlchemy, microservices, event-driven & distributed systems
Databases & Messaging	PostgreSQL, MongoDB, Redis, vector databases (pgvector, Qdrant), RabbitMQ, Celery, queue & message processing
AI / LLM Engineering	LLM integration (OpenAI, Anthropic SDKs), RAG, vector & semantic search, AI agents & multi-agent orchestration, MCP tool servers, prompt engineering, token / cost optimization, streaming
Frontend	React, Next.js, TypeScript, JavaScript, CSS, TailwindCSS, responsive UI/UX, SSR / ISR, accessibility (WCAG, ARIA)
DevOps & Cloud	Azure, GCP, AWS, Docker, Kubernetes (AKS), CI/CD (Azure DevOps, GitHub Actions), Terraform, monitoring & observability
Security	OAuth2 / OIDC, JWT, SSO, Azure AD (EntraID), managed identities, zero-trust, encryption, access control, RBAC
Testing & Practices	Pytest, automated testing, quality gates, Agile / Scrum, code review, mentoring, technical leadership

PROFESSIONAL EXPERIENCE

Belfius — AI Systems Engineer

Jul 2025 – Present • Brussels

Building production-ready, secure, and compliant AI systems within the constraints of a regulated bank.

- Designed RAG retrieval systems enabling LLMs to access internal bank knowledge safely, using vector databases and hybrid search.
- Built secured FastAPI services connecting AI models to enterprise data while enforcing strict access controls.
- Set up Azure infrastructure (AKS, Azure DevOps pipelines) with managed identities, encryption, and zero-trust security policies, aligned with banking standards.
- Developed AI agents that automate internal operations previously requiring manual work, in close collaboration with Architecture teams.

Cohabs — Senior Software Engineer

Apr 2025 – Jun 2025 • Brussels

Scalable payments service for finance, accounting, and payment consolidation.

- Built a payments service handling complex edge cases, mirroring transactions into the finance department's accounting and consolidation tooling.
- Integrated third-party accounting APIs for reliable cross-system communication.
- Enhanced event-driven workflows with robust error handling and queue processing.
- Managed migrations between evolving services with a focus on scalability, traceability, and reliability.

Greenomy — Senior Software Engineer

Jan 2024 – Apr 2025 · Brussels

Scalable SaaS platform delivery and engineering-workflow improvement.

- Automated CI/CD pipelines on Azure DevOps, improving deployment speed and reliability.
- Introduced improved development workflows, monitoring tools, and debugging strategies.
- Mentored engineers, reducing friction in problem-solving and improving team productivity.
- Delivered business-driven features with auto-scalable solutions and long-term stability in mind.

Qover — Senior Software Engineer

2017 – 2024 · Brussels

Early technical team member; helped scale an InsurTech from 13 people to a 150+ employee scale-up.

- Built authentication, third-party integrations, pricing models, claims automation, and compliance systems across all major platform areas (Node.js / NestJS microservices).
- Developed white-label insurance solutions enabling new partners to be onboarded and launched at scale.
- Created contract automation, bordereau generation, financial reconciliation, and RabbitMQ-driven notification workflows.
- Helped migrate and optimize cloud infrastructure from AWS to GCP, ensuring high availability of a platform still running at scale today.

Emakina — Software Engineering Intern

2017 · Brussels

- Developed a digital signage platform for real-time content display across multiple screens.
- Integrated live data feeds (weather, schedules, news) and optimized secure authentication and data-access processes.

SELECTED PROJECTS — PIXL SRL (FOUNDER)

Founder of PIXL SRL — building AI platforms and products, and delivering AI-engineering workshops & enablement for companies.

PIXL — AI Engineering Platform pixldev.be · PyPI

End-to-end platform for AI-native software delivery — orchestration engine, agent crew, RAG knowledge base, and evaluation harness (published to PyPI).

- Built a DAG-based orchestration engine with multi-provider LLMs (Anthropic, OpenAI, Gemini), an event bus, cost tracking, and session resume.
- Shipped a FastAPI service (140+ endpoints, JWT auth, SSE / WebSocket streaming) and a React console with real-time session and DAG visualization.
- Developed a Claude Code crew plugin — 14 specialized agents, 75+ skills, MCP tool servers, and an evaluation harness — over an FTS5 / vector RAG knowledge base.
- Implemented sandboxed execution (Cloudflare Workers), quality gates, RBAC governance, and automated recovery / rollback for AI-generated code.

Feen — AI Accounting Platform feen.be

AI-powered accounting assistant for Belgian SMEs — invoices, transactions, and financial insights in one platform.

- Designed and built the full SaaS architecture with scalable FastAPI backends and data pipelines.
- Developed AI-powered financial document extraction, automatic transaction reconciliation, and LLM data-processing workflows.
- Built real-time financial dashboards and integrated Peppol readiness and company verification.

Python · TypeScript · Next.js · LLM APIs · vector databases · microservices · cloud infrastructure

Pizza Roma — Restaurant POS & Online Ordering Platform pizzeriaroma-anderlecht.be

End-to-end digital platform for a Brussels pizzeria — public site, online ordering, in-store cashier / POS, delivery app, and admin dashboard.

- Built a full-stack Next.js / TypeScript application on PostgreSQL + Prisma with role-based access (admin, cashier, customer, delivery).

- Integrated Stripe payments (PaymentIntent, webhooks, 3D Secure) and transactional email, with a persistent cart and bilingual (FR / NL) storefront.
- Delivered in-store cashier / POS and order-management surfaces, a delivery-driver app, and a real-time admin dashboard.

EDUCATION

Institut Paul Lambin, Brussels — Bachelor in Computer Science

2014 – 2017

LANGUAGES

French (Native) | **English** (Professional)